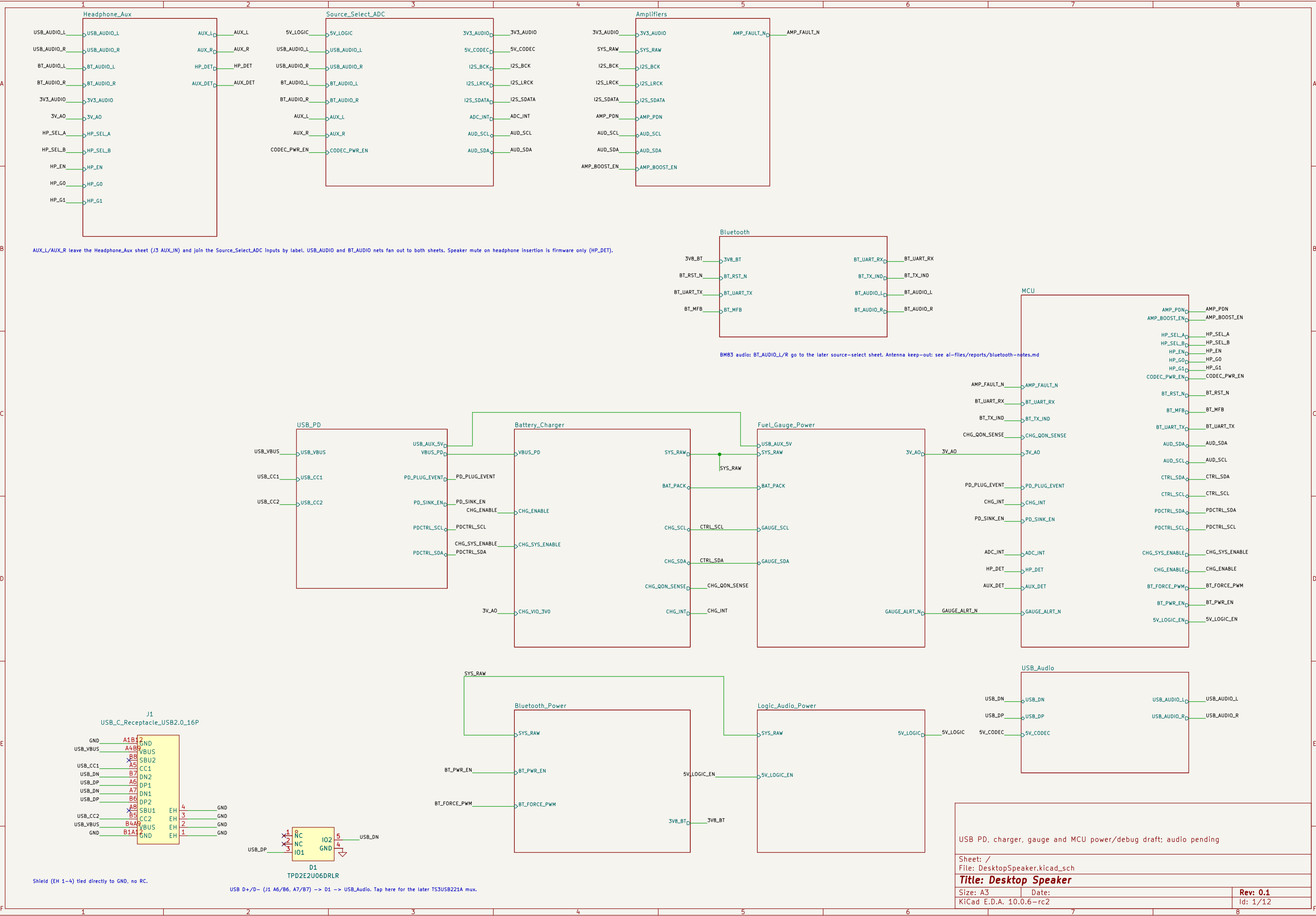
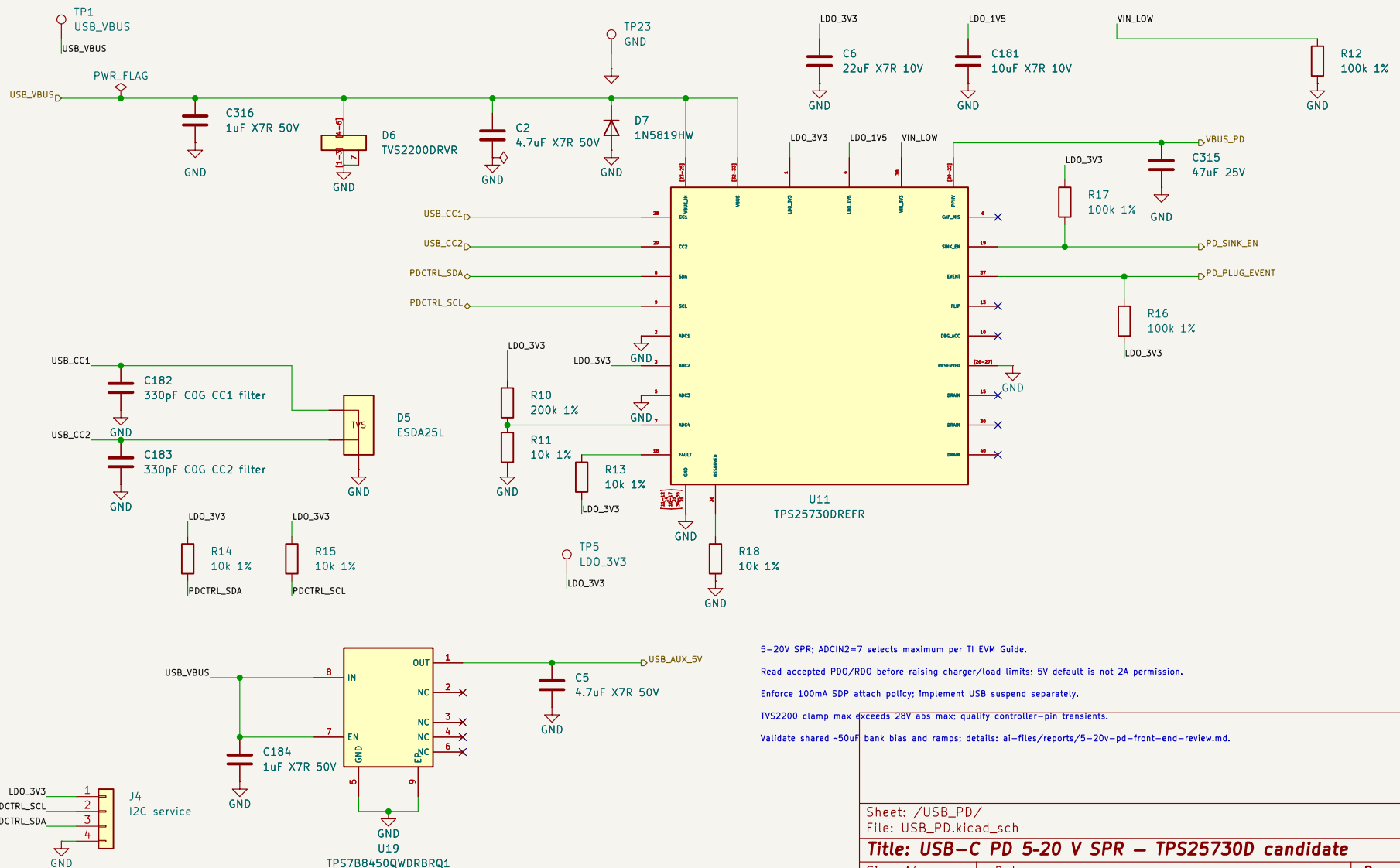






5-20V SPR; ADCIN2=7 selects maximum per TI EVM Guide.

Read accepted PDO/RDO before raising charger/load limits; 5V default is not 2A permission.

Enforce 100mA SDP attach policy; implement USB suspend separately.

TVS2200 clamp max exceeds 28V abs max; qualify controller-pin transients.

Validate shared -50uF bank bias and ramps; details: ai-files/reports/5-20v-pd-front-end-review.md.

Sheet: /USB_PD/

File: USB_PD.kicad_sch

Title: USB-C PD 5-20 V SPR - TPS25730D candidate

Size: A4

Date:

KiCad E.D.A. 10.0.6-rc2

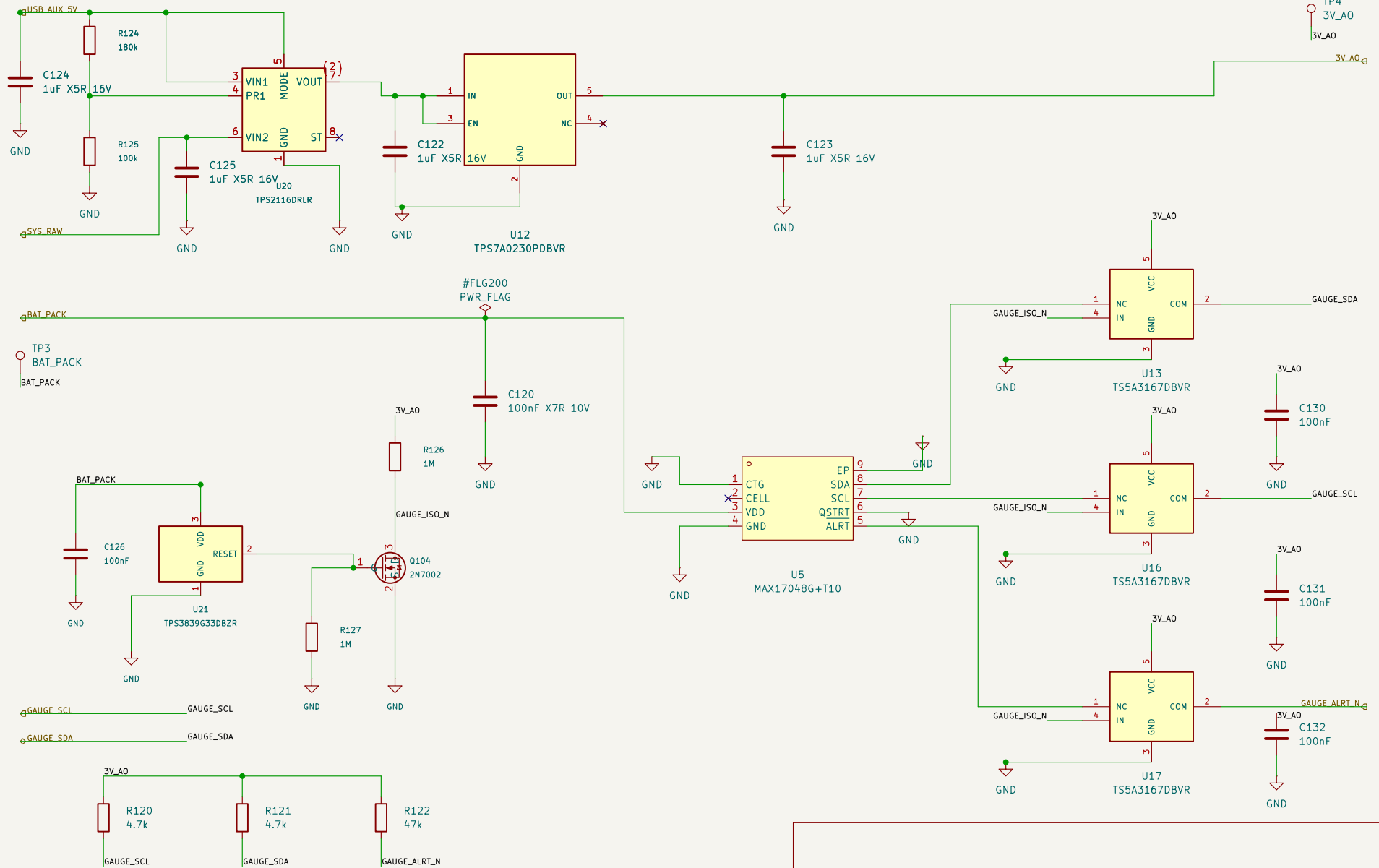
Rev: prototype review

Id: 2/12



I2C pull-ups (10k to logic rail) and CHG_VIO_3V0 level are host-side; D+/D- unconnected (no-connect); set AUTO_INDET_EN=0, HVDCP_EN=0.

Sheet: /Battery_Charger/	
File: Battery_Charger.kicad_sch	
Title: BQ25792 1S charger and power path candidate	
Size: A3	Date:
KiCad E.D.A. 10.0.6-rc2	Rev: candidate-0.1 Id: 3/12



USB-priority logic supply; PR1 nominal 2.8 V.
Switchover/BOR and effective capacitance need qualification.

CELL pad 2 unused on MAX17048; QSTRT tied low

Gauge bus; AO-powered switches; default OFF below valid battery.
U21 3.08 V supervisor enables Q104 after reset recovery.

Sheet: /FuelGauge_Power/
File: FuelGauge_Power.kicad_sch

Title: Fuel gauge and low-current power

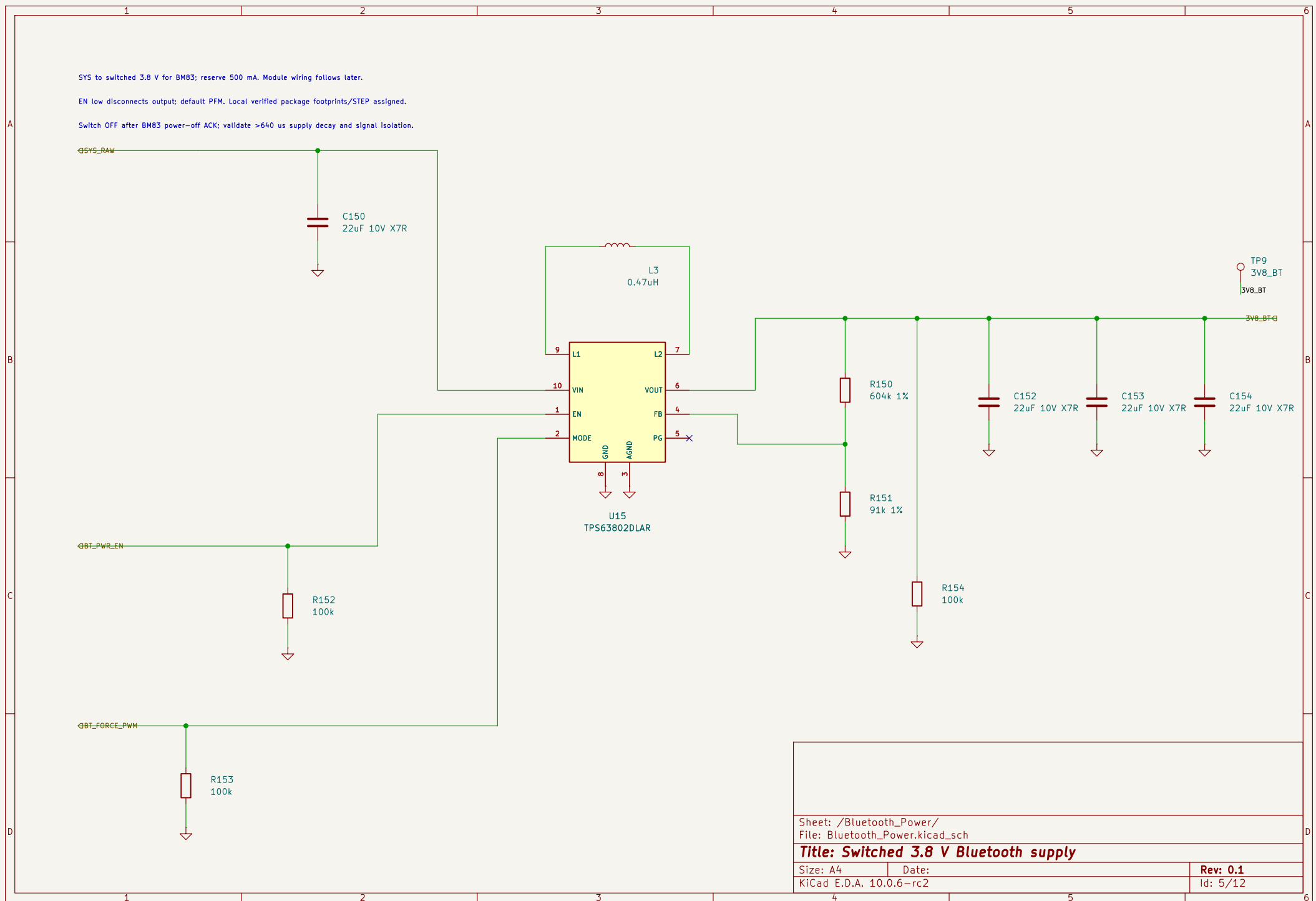
Size: A4

Date:

KiCad E.D.A. 10.0.6-rc2

Rev: 0.1

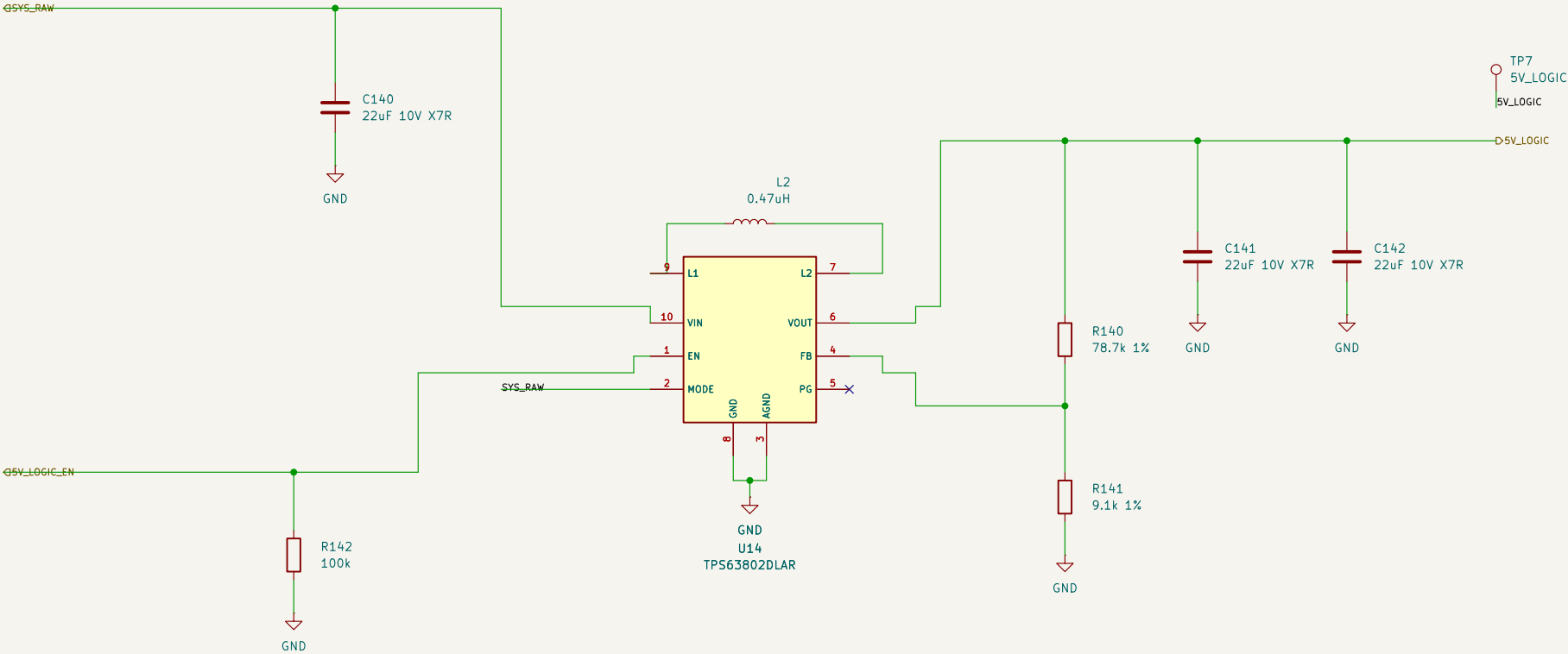
Id: 4/12



Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

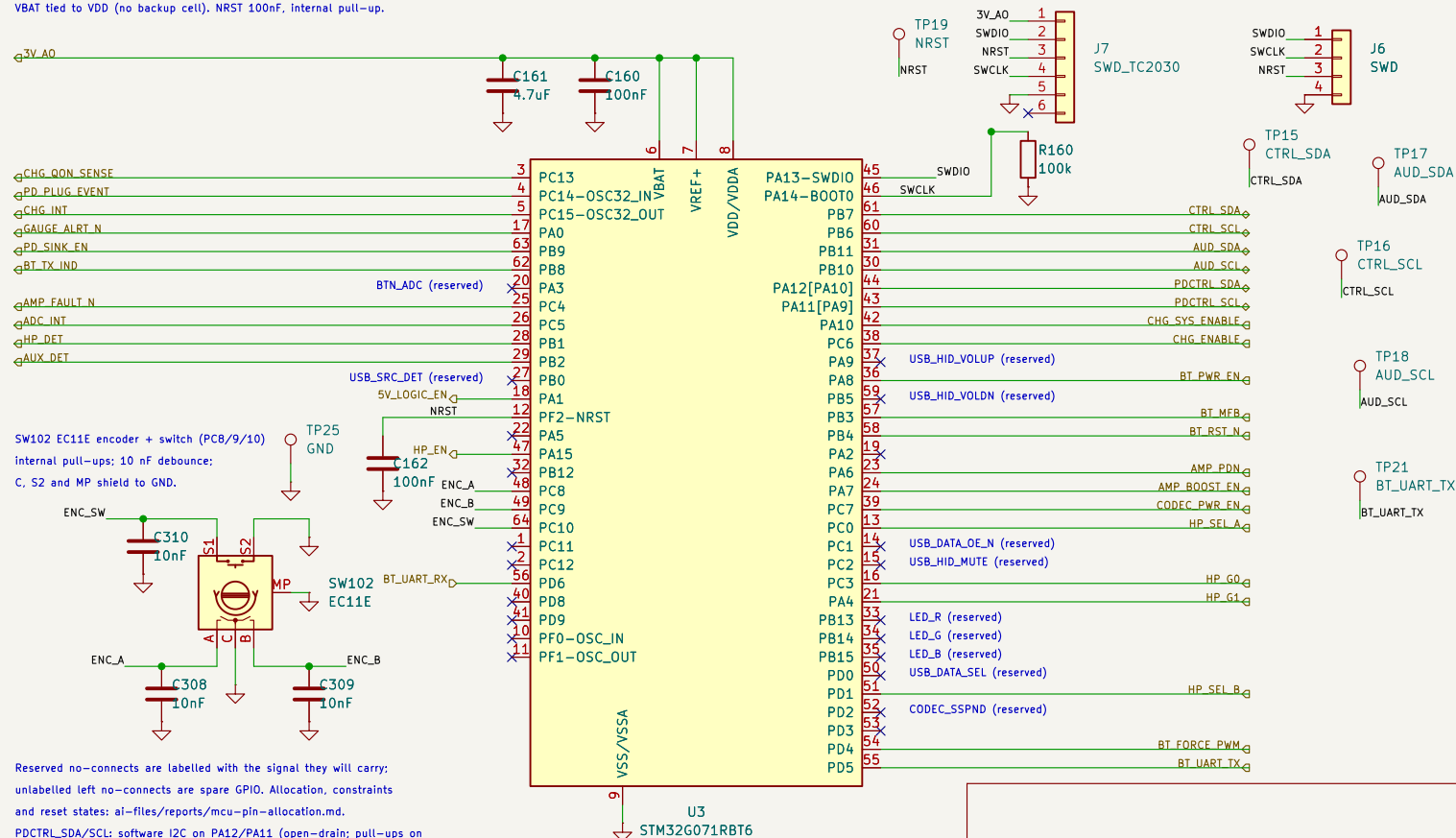
EN low = true input/output disconnect; MODE tied to SYS_RAW forces PWM after soft-start.

Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.



3V_A0 supply: VDD/VDDA 100nF + 4.7uF (DS12232 fig. 13). VREF+ tied to VDDA, VREFBUF off (no separate VREF+ caps).
VBAT tied to VDD (no backup cell), NRST 100nF, internal pull-up.

SWD: J7 Tag-Connect TC2030-IDC (with legs: 6 SWO unused) and J6 4-pin 2.54 mm header share SWDIO, SWCLK, NRST
R160 holds PA14/BOOT0 low (nBOOT_SEL=1 ignores the pin)



Reserved no-connects are labelled with the signal they will carry;
unlabelled left no-connects are spare GPIO. Allocation, constraints
and reset states: ai-files/reports/mcu-pin-allocation.md.
PDCTRL_SDA/SCL: software I2C on PA12/PA11 (open-drain; pull-ups on
the PD sheet), AUD_SDA/SCL: hardware I2C2, PB11/PB10 AF6.
CHG_QON_SENSE: PC13 (WKUP2), input only, high-impedance QON tap.
BT_RST_N: drive open-drain (low only), BT_MFB/BT_UART_TX: drive low
before BT_PWR_EN goes low (3V8_BT back-power).
BT_TX_JND = module PD_0 (EXTI/wake).

STM32G071RBT6 LQFP64: supply, reset, SWD, power-block links

Sheet: /MCU/
File: MCU.kicad_sch

Title: MCU and programming

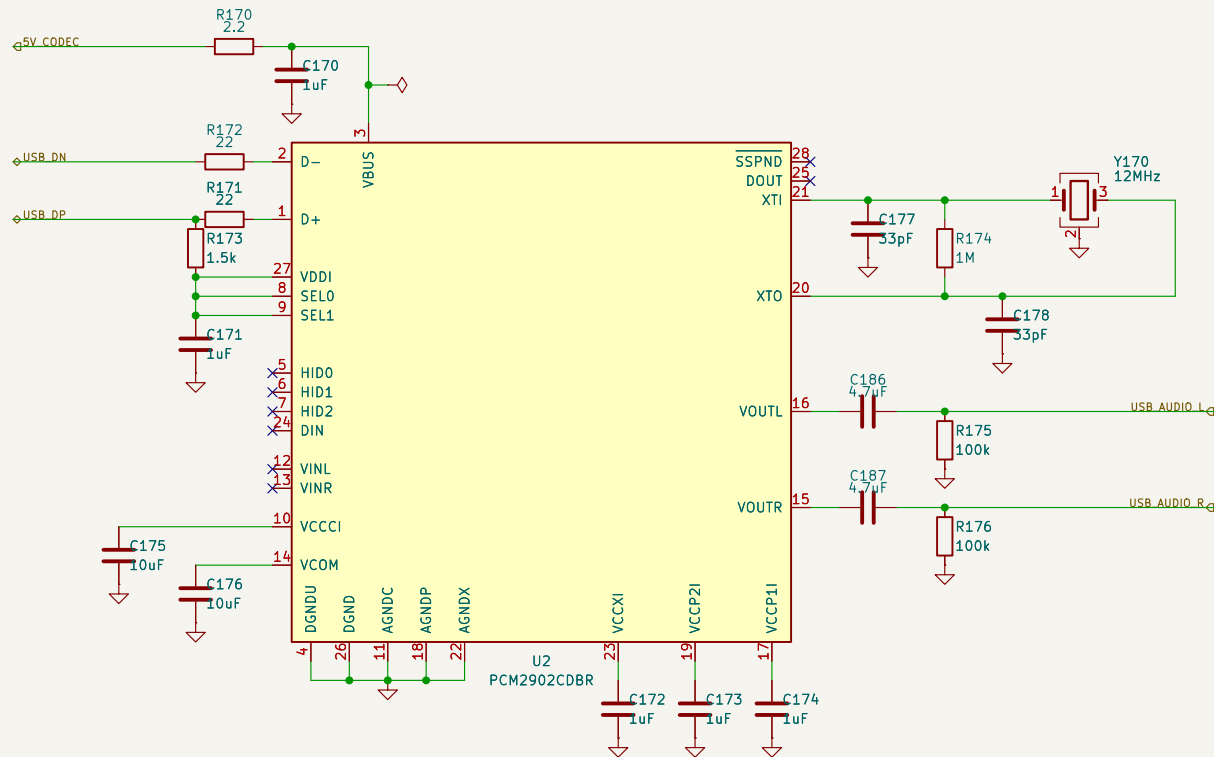
Size: A4
KiCad E.D.A. 10.0.6-rc2

Date:

Rev: 0.1

Id: 7/12

Supply: 5V_CODEC (U23 load switch from 5V_LOGIC, 4.60–5.06 V worst case) feeds VBUS through 2.2R/1uF. DS range 4.35–5.25 V, 56 mA typ / 67 mA max, 250 uA suspend.
 Codec is powered only when CODEC_PWR_EN (U23) and 5V_LOGIC are on: D+ pull-up (R173 to VDDI) = USB attach. Do not enable CODEC_PWR_EN on DCP/unknown sources.
 SEL0/SEL1 = high (VDDI). HID0–2, DIN, VINL/R unused. SSPND (suspend flag, active low) reserved for a later MCU input.
 USB_DP/USB_DN come from J1 via D1 (root). Later source-detect mux goes between those root nets and these ports.



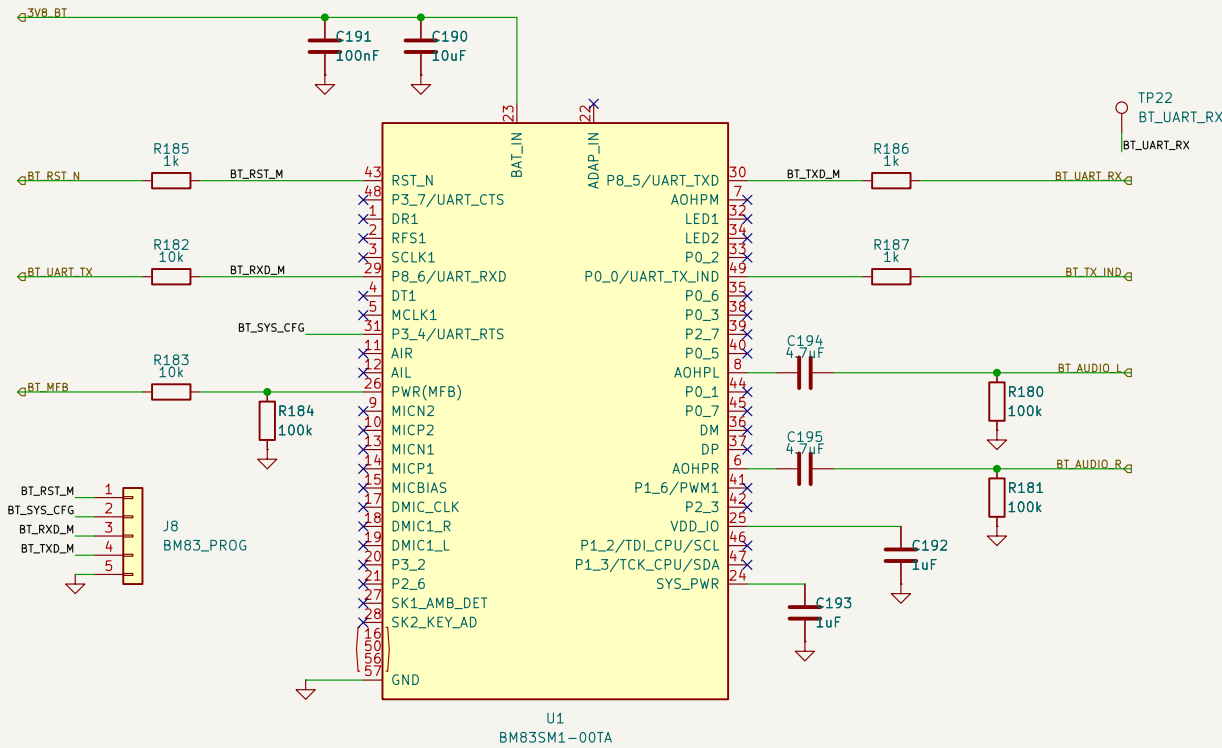
Analogue out: 0.6 VCCCI Vpp (about 2.0 Vpp, 0.7 Vrms) centred on 1.65 V; AC-coupled (HPF -0.4 Hz at 100k), >=10k load.

PCM2902C bus-powered codec from 5V_LOGIC; USB D+/D- path; analogue out to USB_AUDIO_L/R

Sheet: /USB_Audio/
 File: USB_Audio.kicad_sch

Title: USB audio

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 8/12



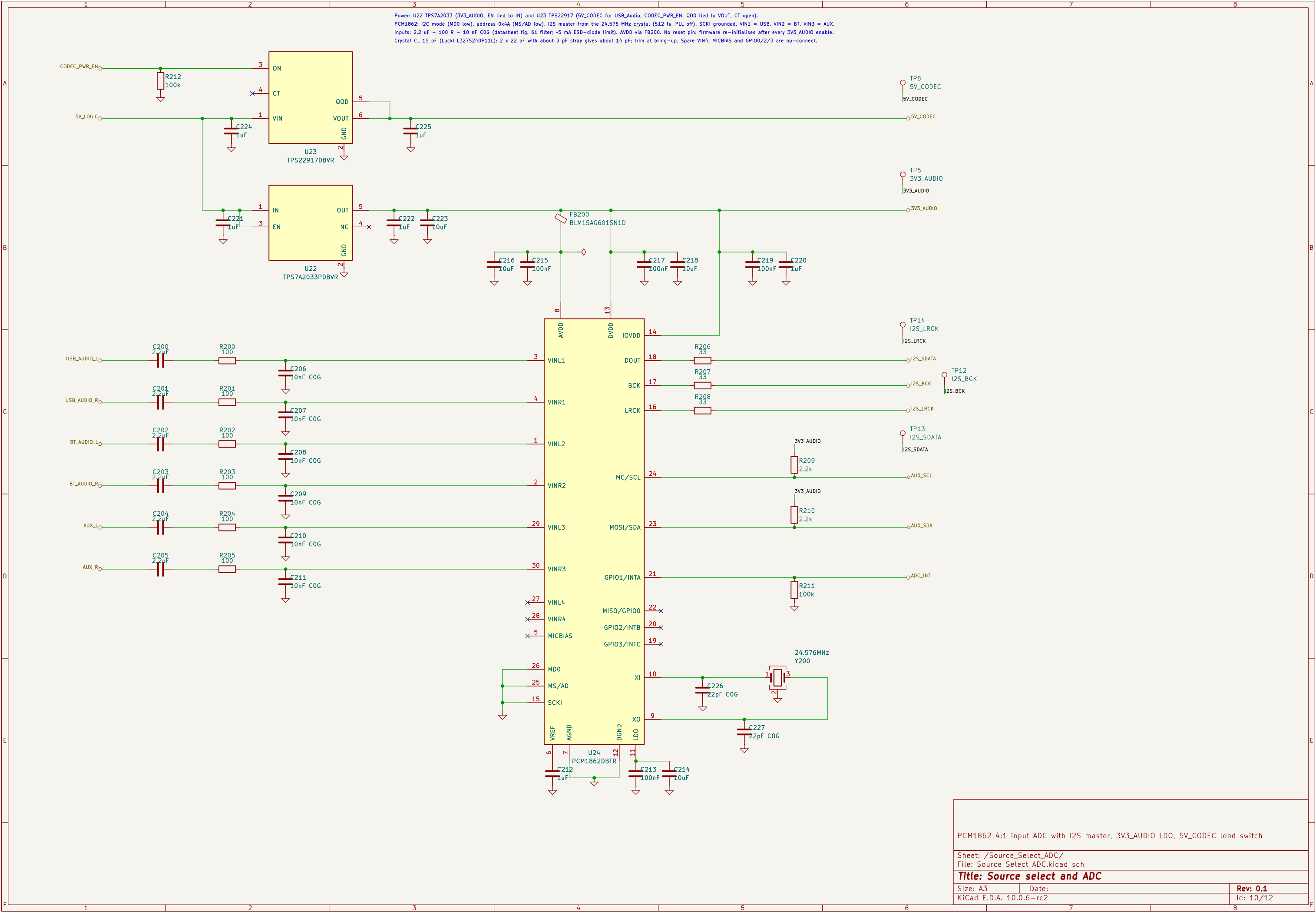
BM83SM1-00TA (DS70005402D). BAT_IN 3.2-4.2 V from 3V8_BT (U15); SYS_PWR/VDD_IO are outputs (decoupling only, do not load or feed). ADAP_IN unused (Internal charger/USB DFU not used).
Host mode UART from the MCU: PB_6 RXD / PB_5 TXD, P0_0 = UART_TX_IND (wakes the MCU). MFB (PWR) = power-on/wake key. RST_N = reset (active low, internal pull-up). Config Tool must set Host mode, P0_0 as TX_IND.
Back-power: MCU lines pass 10k/1k series resistors; firmware must drive PA2/PB3 low (PB4 open-drain/hi-Z) and disable BT_PWR_EN only after the module power-off ACK. Module outputs are 3.0-3.6 V into 5 V tolerant MCU pins.
Audio: single-ended DAC out (AOHPL/AOHPR) AC-coupled, 4.7 uF + 100k bleed, 0.74 Vrms full scale; AOHPM (capless sense) unused. Ports feed the later source-select sheet.
Antenna: integrated PCB antenna, no external RF parts. Keep the antenna end of the module off the board (no copper/components/keep-out) and metal 15 mm clear; see ai-files/reports/bluetooth-notes.md.
J8 programming/test (isUpdate/Config Tool): 1 RST_N, 2 P3_4 SYS_CFG (low at reset = test mode), 3 UART_RXD (adapter TX), 4 UART_TXD (adapter RX), 5 GND. 3V8_BT must be enabled first.

BM83 module: host UART, MFB/RST_N, analogue out

Sheet: /Bluetooth/
File: Bluetooth.kicad_sch

Title: Bluetooth audio

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 9/12

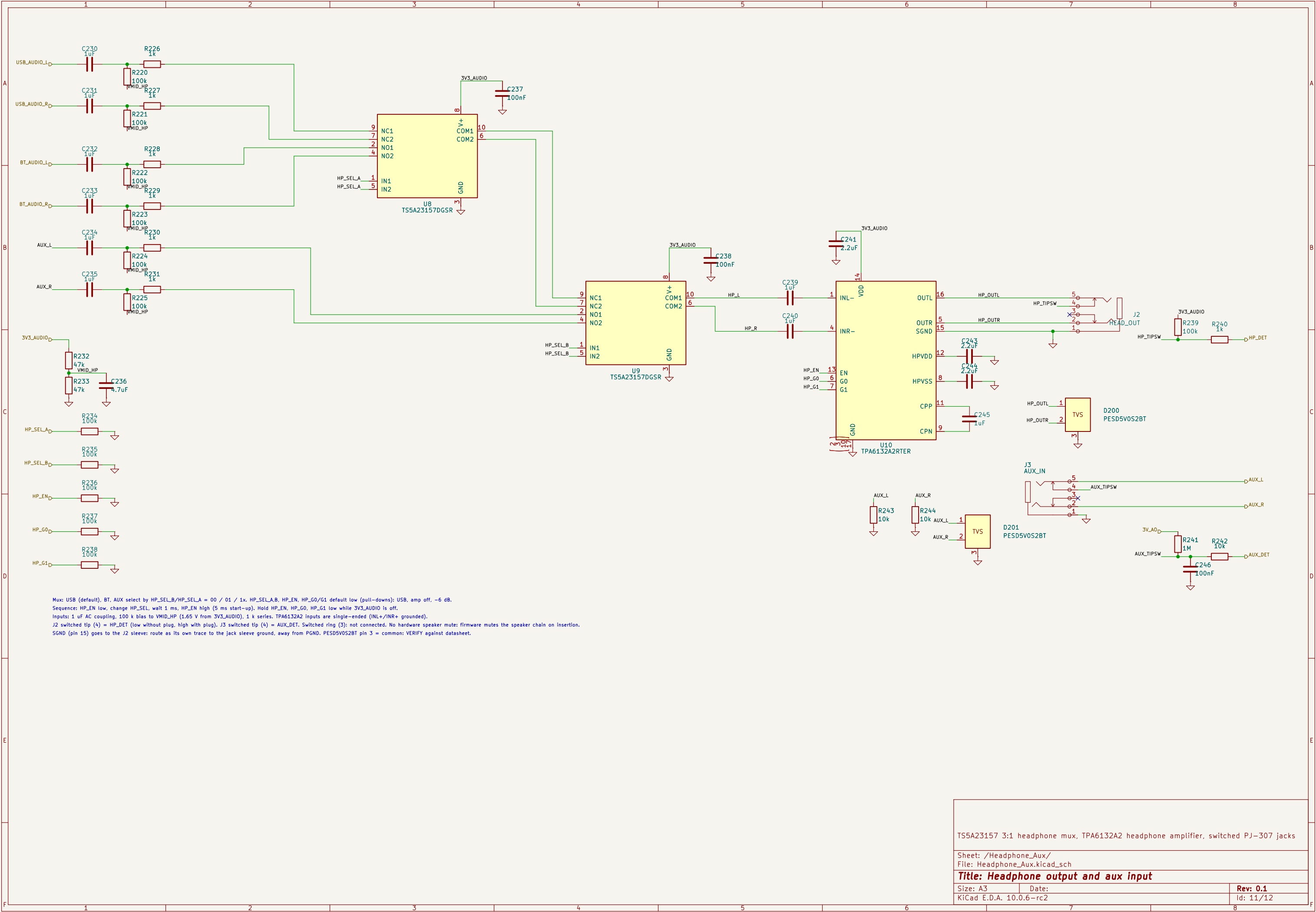


PCM1862 4:1 input ADC with I2S master, 3V3_AUDIO LDO, 5V_CODEC load switch

Sheet: /Source_Select_ADC/
File: Source_Select_ADC.kicad_sch

Title: Source select and ADC

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 10/12



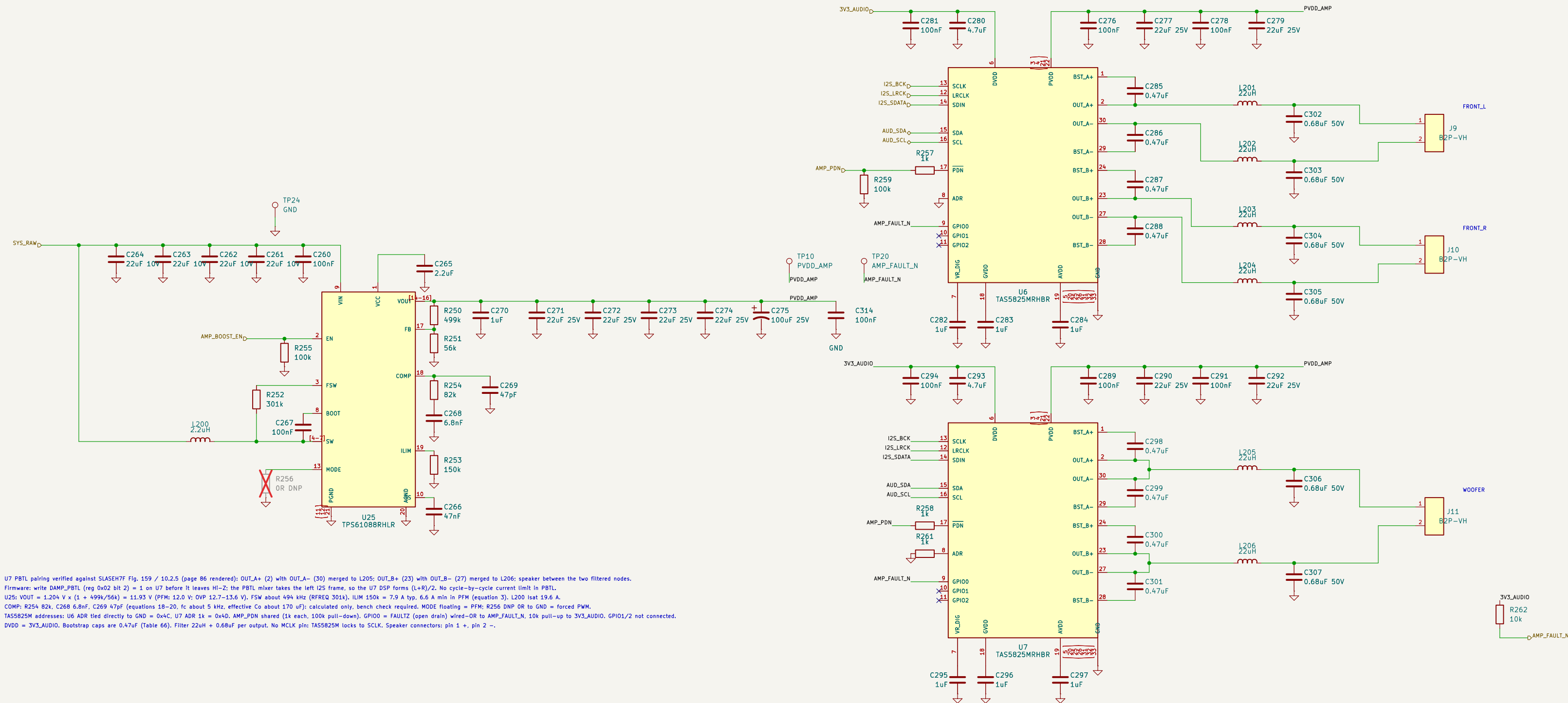
Mux: USB (default), BT, AUX select by HP_SEL_B/HP_SEL_A = 00 / 01 / 1x. HP_SEL_A,B, HP_EN, HP_G0/G1 default low (pull-downs): USB, amp off, -6 dB.
Sequence: HP_EN low, change HP_SEL, wait 1 ms, HP_EN high (5 ms start-up). Hold HP_EN, HP_G0, HP_G1 low while 3V3_AUDIO is off.
Inputs: 1 uF AC coupling, 100 k bias to VMID_HP (1.65 V from 3V3_AUDIO), 1 k series. TPA6132A2 inputs are single-ended (INL+/INR+ grounded).
J2 switched tip (4) = HP_DET (low without plug, high with plug). J3 switched tip (4) = AUX_DET. Switched ring (3): not connected. No hardware speaker mute: firmware mutes the speaker chain on insertion.
SGND (pin 15) goes to the J2 sleeve: route as its own trace to the jack sleeve ground, away from PGND. PESD5V0S2BT pin 3 = common: VERIFY against datasheet.

TS5A23157 3:1 headphone mux, TPA6132A2 headphone amplifier, switched PJ-307 jacks

Sheet: /Headphone_Aux/
File: Headphone_Aux.kicad_sch

Title: Headphone output and aux input

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 11/12



U7 PBTl pairing verified against SLASEH7F Fig. 159 / 10.2.5 (page 86 rendered): OUT_A+ (2) with OUT_A- (30) merged to L205; OUT_B+ (23) with OUT_B- (27) merged to L206; speaker between the two filtered nodes.
Firmware: write DAMP_PBTl (reg 0x02 bit 2) = 1 on U7 before it leaves HI-Z; the PBTl mixer takes the left I2S frame, so the U7 DSP forms (L+R)/2. No cycle-by-cycle current limit in PBTl.
U25: VOUT = $1.204 \text{ V} \times (1 + 499\text{k}/50\text{k}) = 11.93 \text{ V}$ (PFM: 12.0 V; OVP 12.7-13.6 V); FSW about 49k kHz (RFREQ 301k); ILIM 150k = 7.9 A typ, 6.6 A min in PFM (equation 3); L200 isat 19.6 A.
COMP: R254 82k, C268 6.8nF, C269 47pF (equations 18-20, fc about 5 kHz; effective Co about 170 uF); calculated only; bench check required; MODE floating = PFM; R256 DNP OR to GND = forced PWM.
TAS5825M addresses: U6 ADR tied directly to GND = 0x4C; U7 ADR 1k = 0x4D; AMP_PDN shared (1k each, 150k pull-down); GPIO0 = FAULTZ (open drain) wired-OR to AMP_FAULT_N; 10k pull-up to 3V3_AUDIO; GPIO1/2 not connected.
DVDD = 3V3_AUDIO. Bootstrap caps are 0.47uF (Table 66). Filter 22uH + 0.68uF per output. No MCLK pin; TAS5825M locks to SCLK. Speaker connectors: pin 1 +, pin 2 -.

TPS61088 PVDD boost, two TAS5825M (stereo BTL and mono PBTl), LC output filters, speaker connect

Sheet: /Amplifiers/ File: Amplifiers.kicad_sch		
Title: Amplifiers		
Size: A2	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 12/12